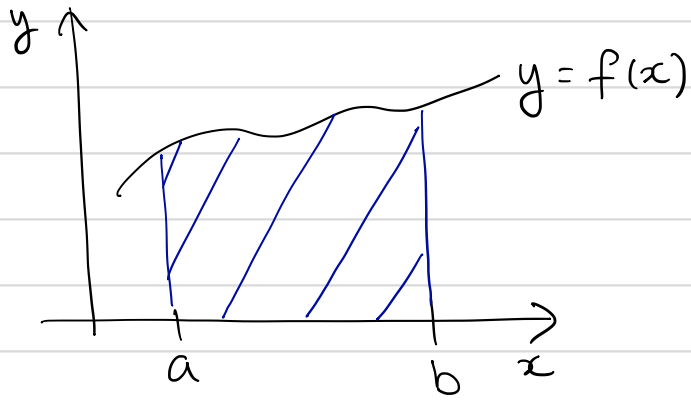
Double Integrals (§15.1-2)

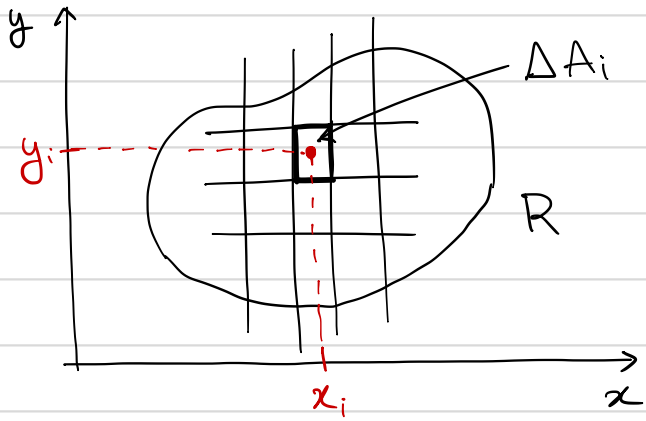
Recall:

$$\int_a^b f(x)\,dx = \text{Area below graph } y=f(x) \text{ over } [a,b]$$

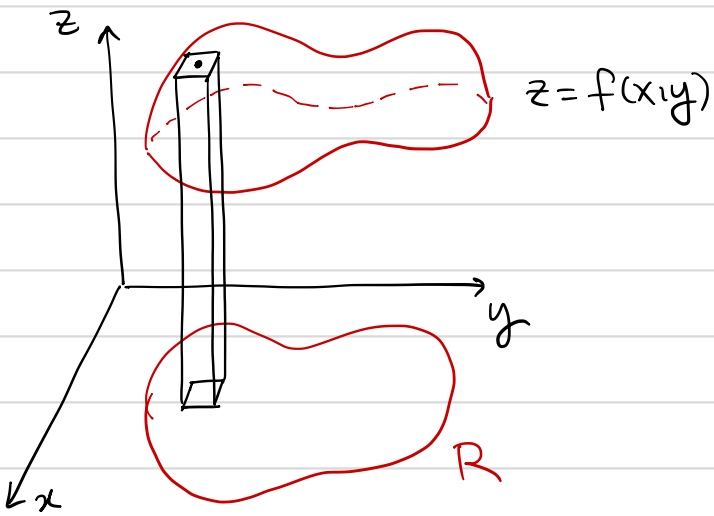


Riemann sums

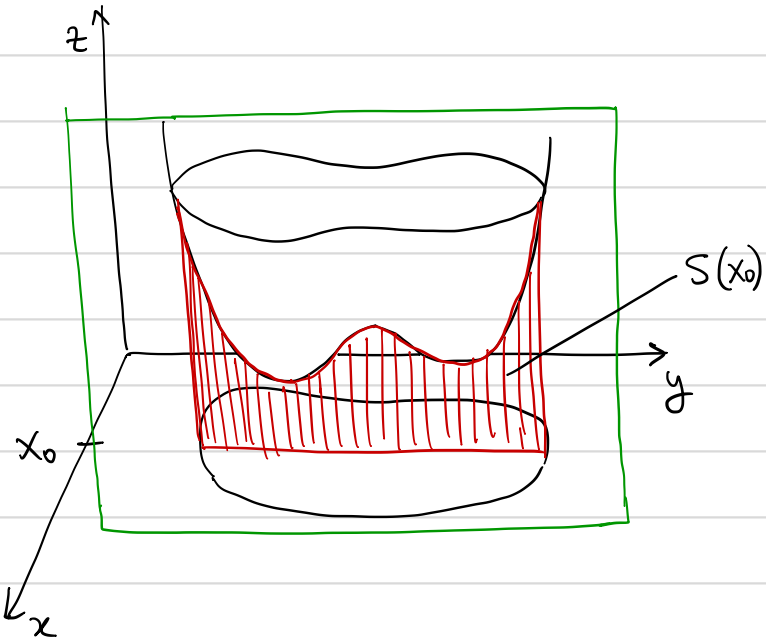
Cut R into small pieces of area ΔA_i.



$$\boxed{\sum_i f(x_i, y_i)\,\Delta A_i \xrightarrow[\Delta A_i \to 0]{} \iint_R f(x,y)\,dA}$$

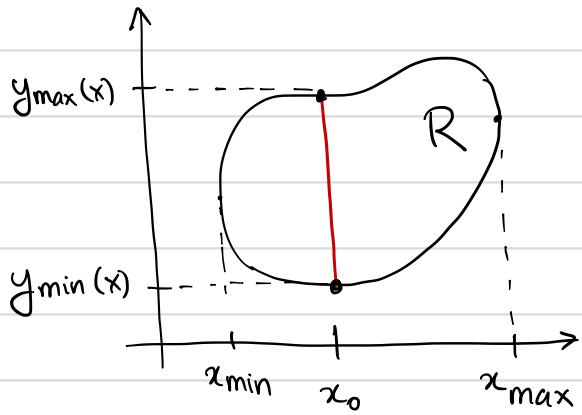


$$\iint_R f(x,y)\,dA = \text{Volume below graph } z=f(x,y) \text{ over region } R \text{ of } xy\text{-plane}$$



Let $S(x_0)$ = area of slice by plane // yz.

Then Volume $= \int_{x_{min}}^{x_{max}} S(x)\,dx$.



For given x,

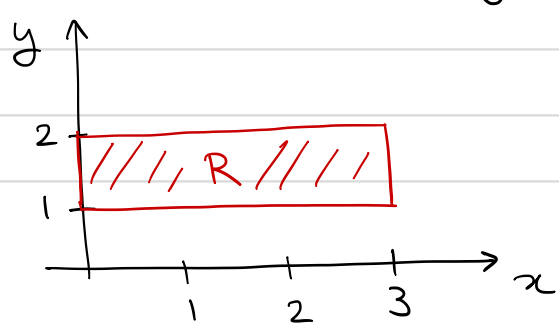
$$S(x) = \int_{y_{min}(x)}^{y_{max}(x)} f(x,y)\,dy$$

Fubini's theorem:

$$\iint_R f(x,y)\,dA = \int_{x_{min}}^{x_{max}} \left[\int_{y_{min}(x)}^{y_{max}(x)} f(x,y)\,dy \right] dx$$

Iterated integral

Example: $f(x,y) = x^2 y$



$R = \{(x,y) \mid 0 \le x \le 3,\ 1 \le y \le 2\}$

$$\iint_R x^2 y\,dA = \int_{x=0}^{3} \left[\int_{y=1}^{2} x^2 y\,dy \right] dx$$

Inner integral:

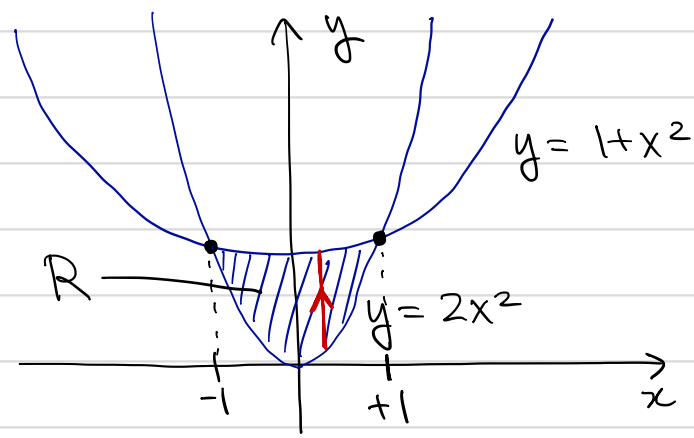
$$\int_1^2 x^2 y\,dy = x^2 \left. \frac{y^2}{2} \right|_{y=1}^{2}$$

$$= x^2 \left(4 - \frac{1}{2}\right) = \frac{3}{2} x^2$$

Outer integral:

$$\int_0^3 \frac{3}{2} x^2\,dx = \frac{3}{2} \left. \frac{x^3}{3} \right|_0^3 = \boxed{\frac{27}{2}}$$

Example: $f(x,y) = x + 2y$



$$\iint_R f(x,y)\,dA = \int_{-1}^{1} \int_{2x^2}^{1+x^2} x + 2y\,dy\,dx$$

$$= \int_{-1}^{1} \left[xy + y^2 \right]_{y=2x^2}^{1+x^2} dx$$

$\rightarrow$

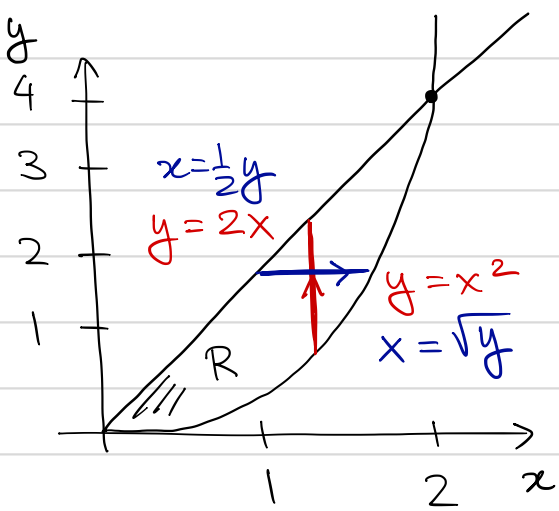
$$= \int_{-1}^{1} \left[x(1+x^2) + (1+x^2)^2 - 2x^3 - 4x^4 \right] dx$$

$$= \int_{-1}^{1} \left[-3x^4 - x^3 + 2x^2 + x + 1 \right] dx$$

$$= -3\frac{x^5}{5} - \frac{x^4}{4} + 2\frac{x^3}{3} + \frac{x^2}{2} + x \Big|_{x=-1}^{1}$$

$$= \frac{32}{15} \quad //$$

<u>Example</u>: $f(x,y) = x^2 + y^2$ (skipped in class)



$$\iint_R f(x,y)\,dA = \int_0^2 \int_{x^2}^{2x} x^2 + y^2 \, dy\, dx$$

$$= \int_0^4 \int_{y/2}^{\sqrt{y}} x^2 + y^2 \, dx\, dy$$

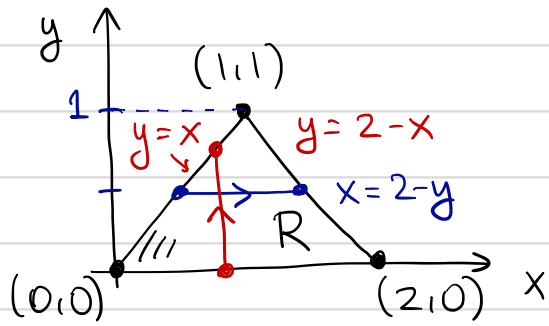
$$= \int_0^4 \left[\frac{x^3}{3} + xy^2 \right]_{x=y/2}^{\sqrt{y}} dy$$

$$= \int_0^4 \left(\frac{y^{3/2}}{3} + y^{5/2} - \frac{y^3}{24} - \frac{y^3}{2} \right) dy$$

$$= \frac{1}{3}\frac{2}{5} y^{5/2} + \frac{2}{7} y^{7/2} - \left(\frac{1}{24} + \frac{1}{2} \right) \frac{y^4}{3} \Big|_0^4$$

$$= \frac{216}{35} \quad //$$

<u>Example</u>: $f(x,y) = e^{x+y}$



$$\iint_R e^{x+y}\, dA$$

$$= \int_{y=0}^{1} \int_{x=y}^{2-y} e^{x+y}\, dx\, dy$$

$$= \int_{y=0}^{1} e^{x+y} \Big|_{x=y}^{2-y} dy$$

$$= \int_0^1 e^2 - e^{2y}\, dy$$

$$= e^2 y - \frac{e^{2y}}{2} \Big|_{y=0}^{1}$$

$$= e^2 - 0 - \frac{e^2}{2} + \frac{1}{2} \quad //$$

The other order:

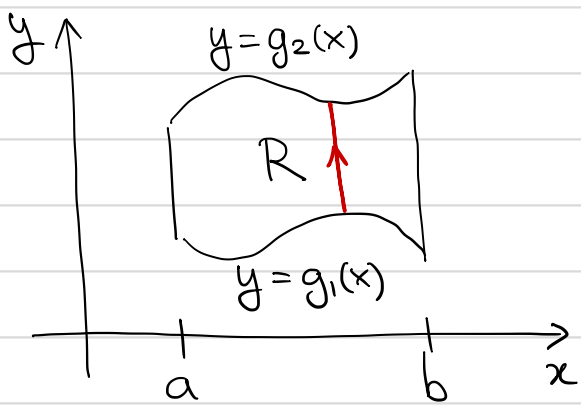
$$\iint_R e^{x+y}\,dA =$$

$$= \int_{x=0}^{1}\int_{y=0}^{x} e^{x+y}\,dy\,dx +$$

$$+ \int_{x=1}^{2}\int_{y=0}^{2-x} e^{x+y}\,dy\,dx$$

(Gives same answer)

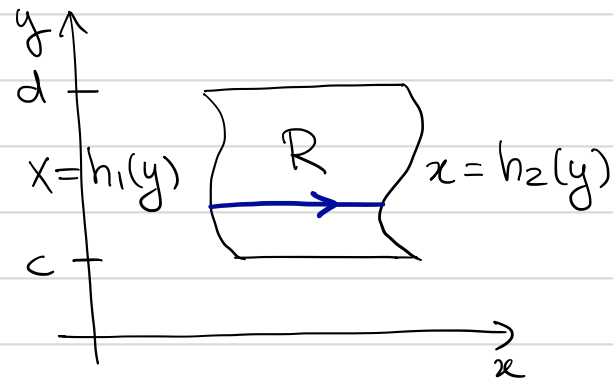
Type I regions (Slides)



$$\iint_R f(x,y)\,dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x,y)\,dy\,dx$$

$$R = \{(x,y) \mid a \leq x \leq b,\ g_1(x) \leq y \leq g_2(x)\}$$

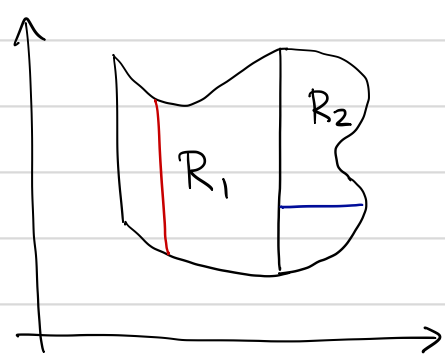
Type II regions (Slides)



$$\iint_R f(x,y)\,dA = \int_c^d \int_{h_1(y)}^{h_2(y)} f(x,y)\,dx\,dy$$

$$R = \{(x,y) \mid c \leq y \leq d,\ h_1(y) \leq x \leq h_2(y)\}$$

Some regions are neither type I nor type II: (Slides)



$$\iint_{R_1 \cup R_2} f(x,y)\,dA = \iint_{R_1} f(x,y)\,dA + \iint_{R_2} f(x,y)\,dA.$$